Freestream	
Alt (kft)	30
M	0.8
P (psf)	628
Pt (psf)	958
T (R)	411.7
Tt (R)	464.4
q (psf)	282
Cdo	0.2
Ao (ft2)	6.36
U (ft/s)	797
md (sl/s)	4.49
An (ft2)	0.6359
Dn (lbf)	36
Dram (lbf)	3580
phi	0.0
beta	3.762
md H2	0.0000
md O2	1.0462
md N2	3.4438
Mb	28.84
R (ft2/s2)	1724
Cp (ft2/s2)	6033

Inlet	
Π_d	0.960
Pt (psf)	920
Tt (R)	464.4
md	4.49
phi	0.0
beta	3.762
md H2	0.0000
md O2	1.0462
md N2	3.4438
Mb	28.84
R (ft^2/s^2)	1724
Cp (ft^2/s^2)	6033

Fan	
Π_f	2.00
η_f	0.95
Pt (psf)	1839
Tt (R)	571
md	4.49
Pwr (ft-lbf/s)	2.90E+06
phi	0.0
beta	3.76
md H2	0.0000
md O2	1.0462
md N2	3.4438
Mb	28.84
R (ft2/s2)	1724
Cp (ft2/s2)	6033

Station 15	
M	0.20
Pt (psf)	1839
P (psf)	1789
Tt (R)	571
T (R)	567
md (sl/s)	4.49
A (ft2)	10.486
U (ft/s)	234
I (lbf)	19807
Ht (ftlb/s)	1.5E+07
phi	0.0
beta	3.76
md H2	0.0000
md O2	1.0462
md N2	3.4438
Mb	28.84
R (ft2/s2)	1724
Cp (ft2/s2)	6033

Dummy Calc for Ttb	
Tt1	540.0
Pt (psf)	200000
md ch	1.000
phi	2
beta	0.0
Cp_H2	86993
Cp_O2	5437
md H2	0.2000
md O2	0.8000
md N2	0.0000
m_brn H2	0.1000
Mbp	10.00
R (ft2/s2)	4971
Cp (ft2/s2)	17399
η_b	59.6%
Ttb	5129

Turbine Exit	
η_τ	0.90
η_μ	0.99
Mx	1.00
Px	1789
Ptx	3386
Tx	1627
Ttx	1952
dh	5.53E+07
md	0.053
phi	2.0
beta	0.0
md H2	0.0106
md O2	0.0424
md N2	0.0000
m_brn H2	0.0053
Mbp	10.00
R (ft2/s2)	4971
Cp (ft2/s2)	17399
Ux	3365
Ax (ft2)	0.0712
I (lbf)	306

Station 7	
md (sl/s)	4.54
phi	0.08
beta	3.615
md H2	0.011
md O2	1.089
md N2	3.444
m_brn H2	0.011
Mb	28.45
R (ft2/s2)	1747
Cp (ft2/s2)	6115
η_b	96.4%
Tt (R)	952
Ht (ftlb/s)	2.6E+07
I (lbf)	20112
A (ft2)	1.1E+01
rho (sl/ft3)	1.05E-03
U (ft/s)	408
T (R)	939
M	0.269
P (psf)	1729
Pt (psf)	1819

Nozzle	
Mb	28.45
R (ft^2/s^2)	1747
Cp (ft^2/s^2)	6115
Tt (R)	952
T (R)	703
Pt (psf)	1819
P (psf)	628
M	1.33
A (ft^2)	5.08
U (ft/s)	1746
md (sl/s)	4.54
Cfg	0.96
F (lbf)	7616

Performance	
FN (lbf)	4000
SFCeng	1.53
FNWA	891
FN/Amx	379
ISP (s)	2346